MIND READER

PROJECT

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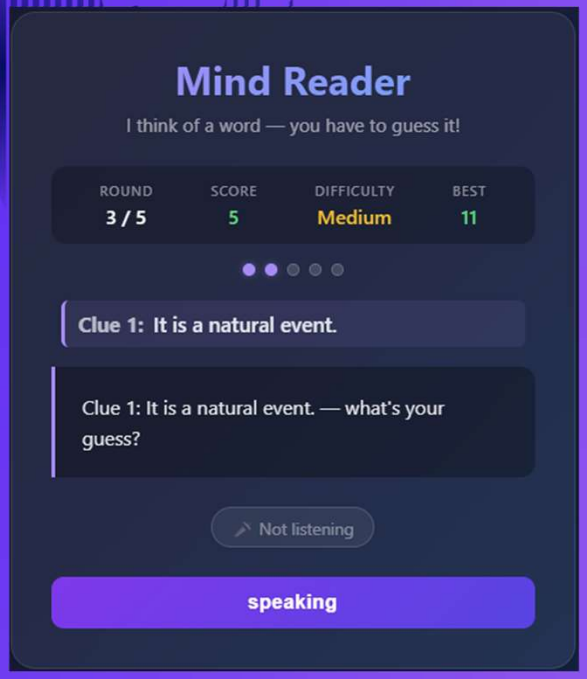
TABLE OF CONTENTS

- The game 03
- Technicalities 04
- Dialogue flow / state machine 05
- Challenges 07
- Relation to Course Contents 08
- Future Work 09

THE GAME

Mind Reader is a voice-only word-guessing game.

- The system thinks of a secret word and gives progressive clues
- The player speaks their guess at any point — no typing, no clicking
- **3 difficulty levels:** Easy, Medium, Hard (60 words total)
- **Voice commands:**
  guess, hint, repeat, skip, give up, help



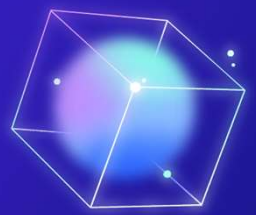
- 5 rounds per game, scored by how early you guess:
  - Clue 1 correct = 5 pts, Clue 2 = 4 pts ... Clue 5 = 1 pt
  - Wrong guess = -1, Skip = -2
  - Win at 10 points, Game Over at -5

TECHNICALITIES

Layer	Technology
Speech recognition & synthesis	Azure Congnitive Services (ASR/TTS)
Natural language understanding	Azure CLU - intent + entity extraction
Dialogue management	XState v5 (hierarchical statecharts)
Speech integration	SpeechState library
Language	TypeScript, running in the browser via Vite
Voice	en-US-DavisNeural

Intents used:

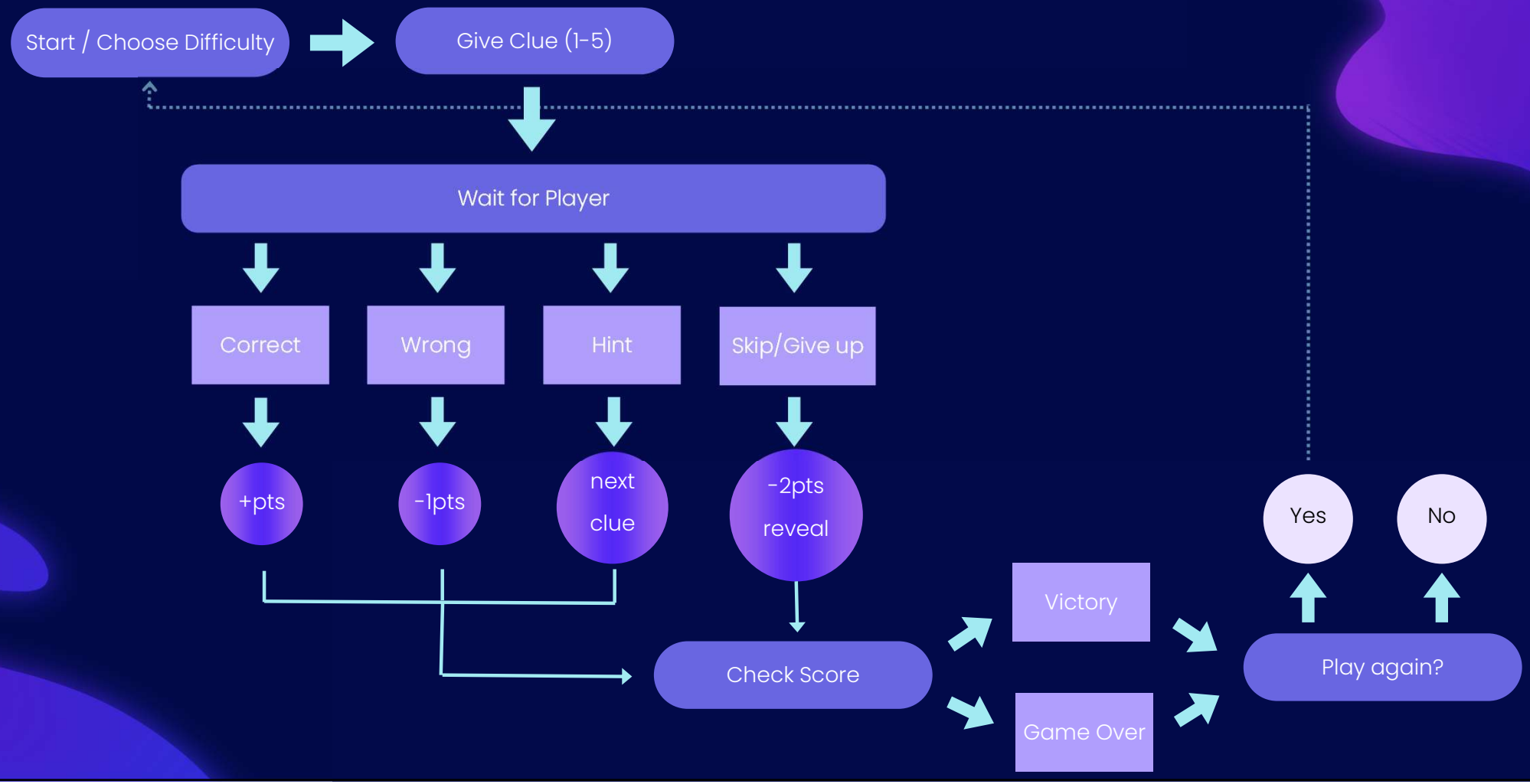
ask_hint
choose_difficulty
give_up
guess_word
help
play_again
repeat_clue
skip_word



Entities:

difficulty
word_guess

DIALOGUE FLOW / STATE MACHINE



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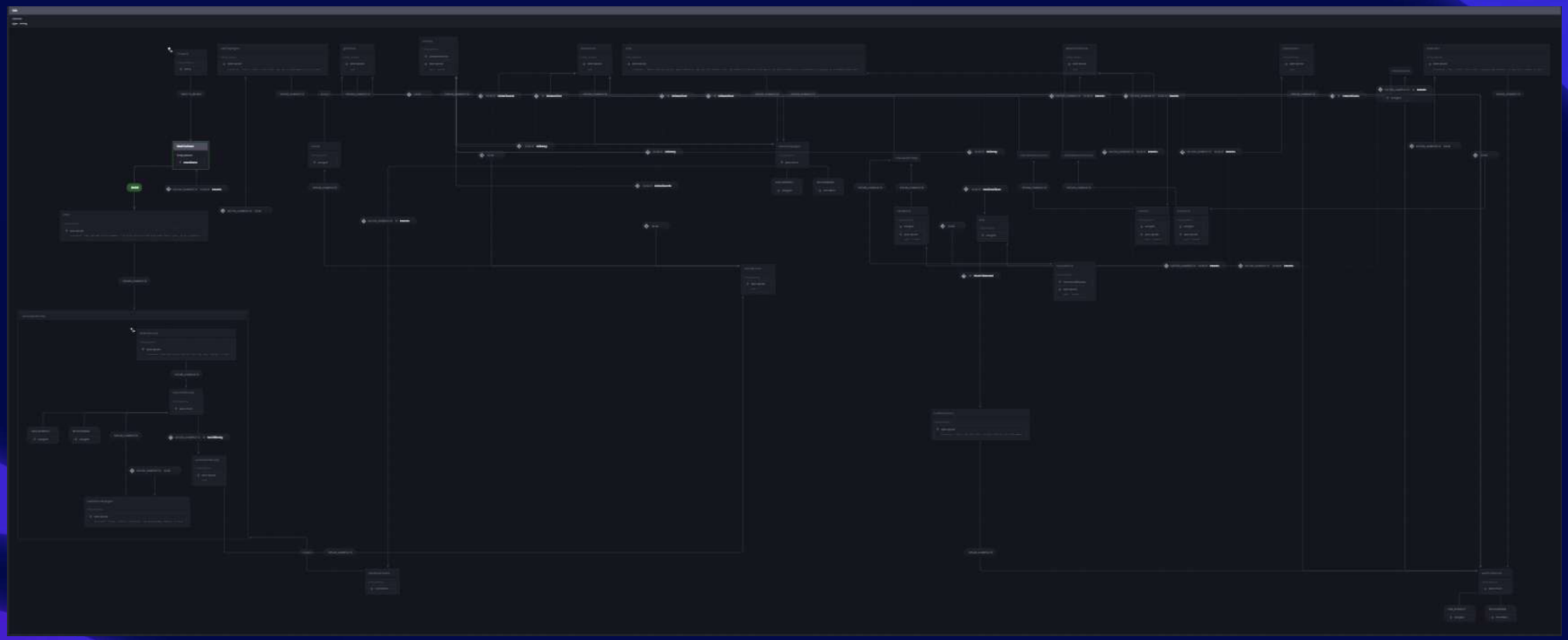


CHALLENGES

ASR/SPEAK RACE CONDITIONS

If you speak text immediately on ASR_NOINPUT or RECOGNISED, the ASR session is still closing and the new speech events get lost or looped.

Solution:
always wait for LISTEN_COMPLETE before transitioning to a speaking state. Used deferEvents: true to queue events during transitions.

SCORE ANNOUNCEMENT TIMING

In XState v5, when you run an assign and a speak action together, the params function already receives the updated context. I was accidentally double-counting the score change, so the spoken score didn't match the UI.

Solution:
he fix was simple: just read context.score directly, since it's already updated by the time params runs.

NLU ENTITY EXTRACTION

Getting the player's guessed word from free speech (e.g. "I think it's banana") required a custom word_guess entity in Azure CLU, not just intent matching.

Course Relation

01 COURSE MATERIALS

- Statecharts / XState
- Dialogue management
- Speech acts
- ASR/TTS pipelines

02 STATECHARTS FIT?

- Yes - discrete states map naturally
- Guards = readable logic
- deferEvents solved concurrency

03 DEVELOPMENT

- Paper diagram first
- Built phase by phase
- Peer feedback added: repeat, score rules, on-screen clues

04 ETHICS

- Voice data sent to Azure - no local option
- Inaccessible for speech impairments
- NLU bias for accents

FUTURE WORK

Conversational:

- Adaptive difficulty based on player performance
- Confidence-based feedback — "I didn't catch that clearly" instead of unknown

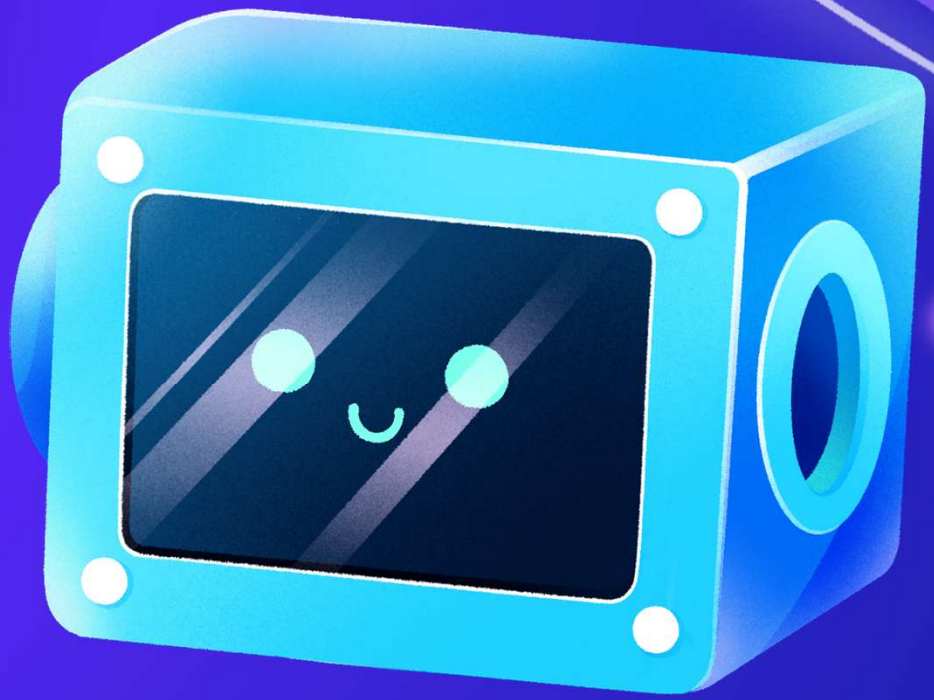
Gameplay:

- Multiplayer mode — one player sets clues, one guesses
- Streak bonuses for consecutive correct answers
- Word categories — animals, science, history...

Technical:

- Expand NLU training data for better accent support
- Text input fallback for accessibility
- Mobile-responsive layout

LIVE DEMO



THANK YOU!